

Generation Interconnection System Impact Study Report

For Queue Project Q13-0200

Carrabelle 345 kV

June 17, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0200. The Interconnection Customer proposes a 239.8 MW wind facility interconnecting at the Carrabelle 345 kV substation in the Virginia area. The request was received on February 01, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 146 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,451,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Siuc - Fairland 138.0 kV: 163.8% loading (181 MVA) following the outage of the Taber - Glenannan 345.0 kV; DFAX = 5.3%.

For the loss of Glenannan Substation - Taber 345.0 kV Line, the Bunker Hill - Beltsville 138.0 kV loads to 163.5% of its 229 MVA rating. Project DFAX: 5.5%.

The Bunker Hill - Nasa 138.0 kV line is overloaded to 142.3% of its 290 MVA emergency rating for the loss of the Glenannan - Taber 345.0 kV Line. The project contributes 5.5% DFAX (13.1 MW) to this overload.

For the loss of Dixleta - Taber 345.0 kV Line, the Drum - Dixleta Substation 138.0 kV loads to 135.9% of its 274 MVA rating. Project DFAX: 6.1%.

Sunbright - Fairland 138.0 kV line: 127.7% loading (276 MVA) following the outage of the Adrian Energy Associates Llc 345/138kV Autotransformer; DFAX = 6.2%.

For the loss of Adrian Energy Associates Llc 345 kV/138 kV TX, the Sunbright - Colesville 138.0 kV line loads to 126.9% of its 322 MVA rating. Project DFAX: 6.2%.

For the loss of Taber - Dixleta 345.0 kV L, the Jerome - High Ridge Substation 138.0 kV line loads to 123.3% of its 252 MVA rating. Project DFAX: 9.0%.

The Sierra Pacific Lumber 138/1kV TX is overloaded to 123.1% of its 528 MVA emergency rating for the loss of the Taber - Glenannan 345.0 kV. The project contributes 7.0% DFAX (16.7 MW) to this overload.

Taber 345/138 kV XFMR: 121.0% loading (551 MVA) following the outage of the Dixleta - Taber 345.0 kV; DFAX = 5.3%.

The Lake Crk - Sierra Pacific Lumber 138.0 kV is overloaded to 120.1% of its 322 MVA emergency rating for the loss of the Glenannan - Taber 345.0 kV L. The project contributes 5.3% DFAX (12.8 MW) to this overload.

For the loss of Tombstone - Bixby Hill 500.0 kV Line, the Sharyland - Bixby Hill 345.0 kV loads to 113.9% of its 1,414 MVA rating. Project DFAX: 7.7%.

For the loss of Taber - Glenannan Substation 345.0 kV line, the Sierra Pacific Lumber 345-1 kV Transformer loads to 103.5% of its 628 MVA rating. Project DFAX: 7.0%.

The Wilkens - Highgrove 138.0 kV L is overloaded to 103.0% of its 259 MVA emergency rating for the loss of the Dixleta - Taber 345.0 kV. The project contributes 5.9% DFAX (14.2 MW) to this overload.

Tombstone Substation - Sharyland 345.0 kV: 100.2% loading (1,653 MVA) following the outage of the Bixby Hill - Tombstone 500.0 kV; DFAX = 7.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fairland - Siuc 138.0 kV	Rebuild line with higher-capacity conductor	\$1,563,000	9% (\$135,000)
Beltsville - Bunker Hill 138.0 kV L	Rebuild line with higher-capacity conductor	\$603,000	7% (\$42,000)
Bunker Hill - Nasa 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,781,000	7% (\$124,000)
Drum - Dixleta Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,979,000	6% (\$162,000)
Sunbright - Fairland 138.0 kV line	Rebuild line with higher-capacity conductor	\$534,000	8% (\$43,000)
Sunbright - Coleville 138.0 kV	Rebuild line with higher-capacity conductor	\$1,301,000	8% (\$105,000)
Jerome - High Ridge 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,142,000	4% (\$87,000)
Sierra Pacific Lumber 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,864,000	4% (\$201,000)
Taber 345-138 kV XFMR	Replace transformer with higher-rated unit	\$9,052,000	5% (\$416,000)
Lake Crk - Sierra Pacific Lumber 138.0 kV	Rebuild line with higher-capacity conductor	\$2,372,000	9% (\$213,000)
Sharyland - Bixby Hill 345.0 kV	Rebuild line with higher-capacity conductor	\$43,254,000	3% (\$1,394,000)
Sierra Pacific Lumber 345-1 kV Transformer	Replace transformer with higher-rated unit	\$22,870,000	4% (\$947,000)
Wilkens - Highgrove 138.0 kV Line	Reconductor line	\$241,000	5% (\$13,000)
Tombstne - Sharyland 345.0 kV	Reconductor line	\$17,697,000	3% (\$569,000)
	Total allocated to project		\$4,451,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.